Time series analysis

AB–204 Arctic population ecology, October 27–30, 2026

J.-Prof. Dr. Timo Adam

Course overview

Visualising (time series) data
October 27, 2026

- what is a time series?
- R and RStudio
- scatter and time series plots
- boxplots, histograms, densities

2 Basics of linear regression
October 28, 2026

- statistical models
- simple and multiple linear regression
- interpreting effects
- residuals and model selection

Trend, seasonal, and cyclical effects
October 29, 2026

- time series components
- trend and seasonality
- autocorrelation (ACF, PACF)
- autoregressive models

Summary and outlook
October 30, 2026

- summary of the course
- outlook: other extensions of linear regression
- your questions

Questions? menti.com/al94rhvnrre9.

Lecture 2: Basics of linear regression

2.1 Introduction — what is statistical modelling?

2.2 Simple linear regression

2.3 Model checking

2.4 Multiple linear regression

2.5 A non-technical summary

2.1 Introduction — what is a statistical model?

- Ecological data are **noisy**: two penguins with the same flipper length hardly ever have the same bill length.
- Reasons: measurement error, unobserved influences (age, sex, food availability, ...), and the complexity of nature.
- A **statistical model** splits each observation into two parts:

$$\text{observation} = \underbrace{\text{signal}}_{\text{systematic part}} + \underbrace{\text{noise}}_{\text{random part}} .$$

- The signal is what we're interested in (e.g., how the number of bikes changes with temperature); the noise is described by a **probability distribution**.
- A model is a **simplified** representation of reality — still useful to quantify relationships, identify patterns, and forecast.

"All models are wrong, but some are useful." (George Box, statistician)

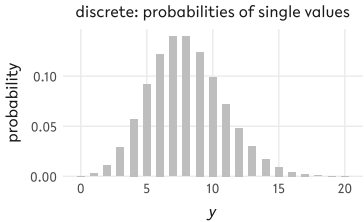
2.1 Introduction — random variables

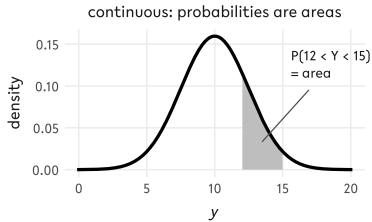
A **random variable** Y is a variable whose value is subject to chance. Its **probability distribution** describes which values are possible, and how likely they are:

- **Discrete**: separate values, often counts (e.g., number of chicks in a nest);
- **Continuous**: any value in an interval (e.g., body mass, temperature, sea ice extent).



Artwork by Allison Horst

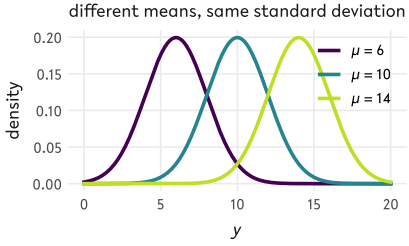


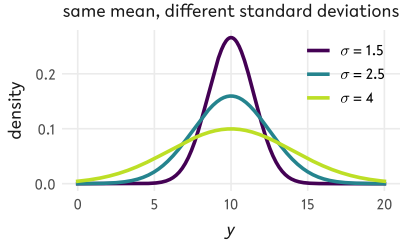


2.1 Introduction — mean and standard deviation

The **normal distribution** $\mathcal{N}(\mu, \sigma^2)$ — bell-shaped and symmetric — is described by two numbers:

- the **mean** μ: the expected value (where the distribution is centred);
- the **standard deviation** σ: the typical deviation from the mean (the variance is σ^2).





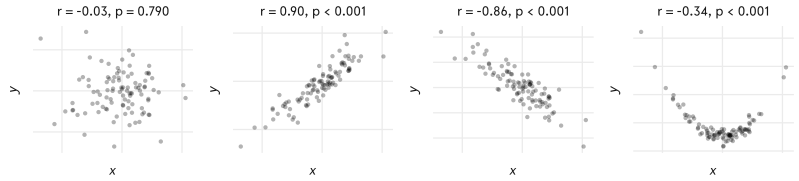
About 95 % of the values lie within $\mu \pm 2 \cdot \sigma$. In R, estimate both from data with `mean(y)` and `sd(y)`.

2.1 Introduction — correlation

The **correlation coefficient** measures the strength of the **linear** relationship between two variables:

$$r_{x,y} = \frac{\sum_{t=1}^{T}(x_t - \bar{x}) \cdot (y_t - \bar{y})}{\sqrt{\sum_{t=1}^{T}(x_t - \bar{x})^2} \cdot \sqrt{\sum_{t=1}^{T}(y_t - \bar{y})^2}} \in [-1, 1],$$

where $\bar{x}$ and $\bar{y}$ are the means of x and y ($r = 0$: no linear relationship).



(simulated data)

```
cor(x, y) # correlation coefficient
cor.test(x, y) # test if the correlation is different from 0 (p-value)
```

Correlation only captures **linear** relationships (right panel) — and it does not imply causation.

Lecture 2: Basics of linear regression

2.1 Introduction — what is statistical modelling?

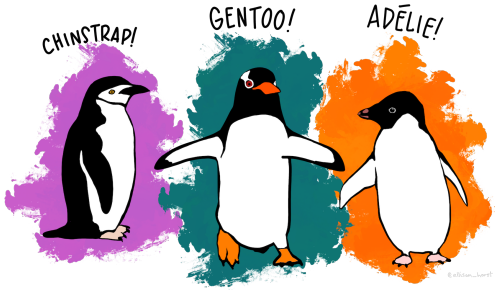
2.2 Simple linear regression

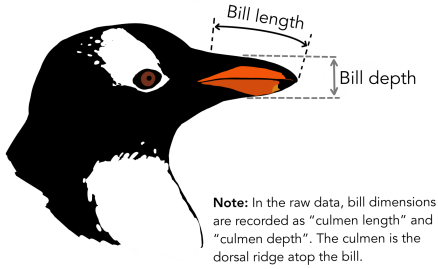
2.3 Model checking

2.4 Multiple linear regression

2.5 A non-technical summary

2.2 Simple linear regression — the Palmer penguins



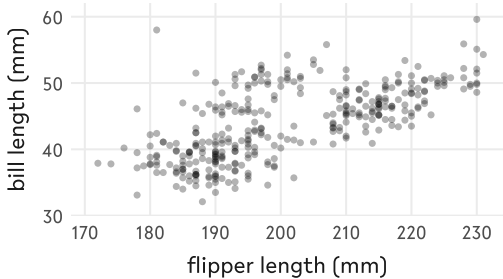


Measurements of 344 penguins of three species from the Palmer Archipelago, Antarctica[1].

[1]Data from Gorman, K., Williams, T., and Fraser, W. (2014): Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLoS ONE*, 9, e90081, doi:10.1371/journal.pone.0090081; available as `penguins` in the `R` package `palmerpenguins`. Artwork by Allison Horst.

2.2 Simple linear regression — the Palmer penguins

```
library(palmerpenguins) # install.packages("palmerpenguins") once
plot(x = penguins$flipper_length_mm, y = penguins$bill_length_mm, pch = 20,
     col = adjustcolor("black", alpha.f = 0.3),
     xlab = "flipper length (mm)", ylab = "bill length (mm)")
```



Can the **bill length** be explained by the **flipper length**?

2.2 Simple linear regression — regression

- **Regression** models explain a **response** Y_t by a **covariate** x_t:

$$Y_t = f(x_t) + \varepsilon_t, \qquad \mathbb{E}(\varepsilon_t) = 0 \qquad \Leftrightarrow \qquad \mathbb{E}(Y_t) = f(x_t),$$

  where f is some function (the signal) and ε_t is an error term (the noise).
- Other names: the response is also called **outcome** or **dependent variable**; the covariate **predictor**, **explanatory** or **independent variable**.
- The idea is to detect the signal f while accounting for the noise ε_t.

(For penguins, the index t counts individuals rather than time points — the methods are the same.)

2.2 Simple linear regression — the model

In **linear** models, the signal f is a straight line:

The **simple linear model** is given by

$$Y_t = \beta_0 + \beta_1 \cdot x_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2) \qquad \Leftrightarrow \qquad \mathbb{E}(Y_t) = \beta_0 + \beta_1 \cdot x_t,$$

where

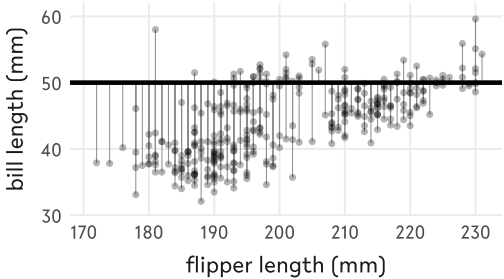
- β_0 is the **intercept** and β_1 the **slope** (the regression coefficients);
- the errors ε_t are independent and normally distributed with mean 0 and variance σ^2.

Assuming linearity makes our life easier — assuming normally distributed errors further simplifies statistical inference.

2.2 Simple linear regression — least squares

Which line fits best? **Least squares**: choose β_0 and β_1 such that the **sum of squared residuals** (vertical distances between the observations and the line) is as small as possible:

$$\text{SSR}(\beta_0, \beta_1) = \sum_{t=1}^{T} \underbrace{\left(y_t - (\beta_0 + \beta_1 \cdot x_t)\right)}_{\text{residual}}^2.$$



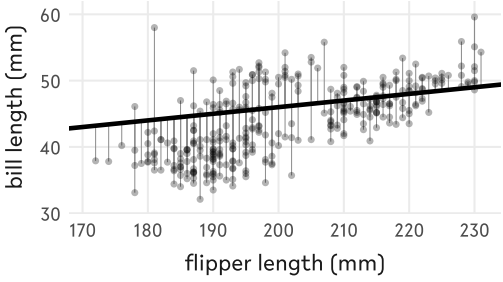
A **poor** fit:

- $\beta_0 = 50$, $\beta_1 = 0$ (a flat line);
- large residuals: $\text{SSR}(50, 0) = 22{,}798.7$.

2.2 Simple linear regression — least squares

Which line fits best? **Least squares**: choose β_0 and β_1 such that the **sum of squared residuals** (vertical distances between the observations and the line) is as small as possible:

$$\text{SSR}(\beta_0, \beta_1) = \sum_{t=1}^{T} \underbrace{\left(y_t - (\beta_0 + \beta_1 \cdot x_t)\right)}_{\text{residual}}^2.$$



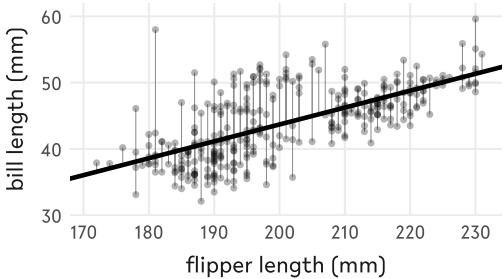
A **better** (but not the best) fit:

- $\beta_0 = 26$, $\beta_1 = 0.1$;
- smaller residuals: $\text{SSR}(26, 0.1) = 9{,}012.7$.

2.2 Simple linear regression — least squares

Which line fits best? **Least squares**: choose β_0 and β_1 such that the **sum of squared residuals** (vertical distances between the observations and the line) is as small as possible:

$$\text{SSR}(\beta_0, \beta_1) = \sum_{t=1}^{T} \underbrace{\left(y_t - (\beta_0 + \beta_1 \cdot x_t)\right)}_{\text{residual}}^2.$$



The **best** fit:

- $\hat{\beta}_0 = -7.265$, $\hat{\beta}_1 = 0.255$ (estimates have a hat);
- no other line has a smaller SSR $= 5{,}787.8$;
- found by differentiating the SSR (details not covered) — R does this for us.

2.2 Simple linear regression — estimation in R

```
mod_1 <- lm(formula = bill_length_mm ~ flipper_length_mm, data = penguins) # fit
summary(mod_1) # show the model output
```

```
Residuals:
    Min      1Q  Median      3Q     Max
-8.5792 -2.6715 -0.5721  2.0148 19.1518

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       -7.26487    3.20016   -2.27   0.0238 *
flipper_length_mm  0.25477    0.01589   16.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

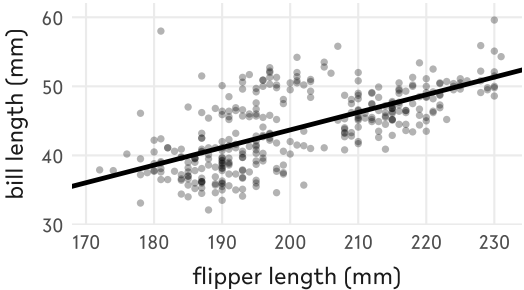
Residual standard error: 4.126 on 340 degrees of freedom
  (2 observations deleted due to missingness)
Multiple R-squared:  0.4306,^^IAdjusted R-squared:  0.4289
```

How to read the output:

- `Estimate`: $\hat{\beta}_0$ and $\hat{\beta}_1$;
- `Std. Error`: their uncertainty;
- `Pr(>|t|)`: p-value (next slides);
- `Residual standard error`: $\hat{\sigma}$;
- `Multiple R-squared`: proportion of the variability explained.

2.2 Simple linear regression — plotting the fitted model

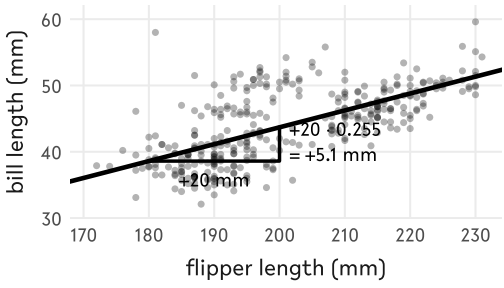
```r
plot(x = penguins$flipper_length_mm, y = penguins$bill_length_mm, pch = 20,
     col = adjustcolor("black", alpha.f = 0.3),
     xlab = "flipper length (mm)", ylab = "bill length (mm)")
abline(a = coef(mod_1)[1], b = coef(mod_1)[2], lwd = 2) # add the line
```



The fitted model:

$$\begin{aligned}\mathbb{E}(\text{bill length}_t) &= \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{flipper length}_t \\ &= -7.265 + 0.255 \cdot \text{flipper length}_t.\end{aligned}$$

2.2 Simple linear regression — interpretation



- **Slope**: if the flipper length increases by 1 mm, the expected bill length increases by $\hat{\beta}_1 = 0.255$ mm (by 20 mm: 5.1 mm).
- **Prediction**: for a flipper length of 200 mm, the expected bill length is

$$-7.265 + 0.255 \cdot 200 = 43.7 \text{ mm}.$$

- **Intercept**: the expected bill length for a flipper length of 0 mm — biologically meaningless.

Be careful with predictions outside the range of the data (**extrapolation**).

2.2 Simple linear regression — is the effect real?

Even without any relationship, $\hat{\beta}_1$ would not be exactly 0, simply due to chance. Is $\hat{\beta}_1 = 0.255$ clearly different from 0?

- The **standard error** quantifies the uncertainty of an estimate: here, 0.016.
- A **95 % confidence interval** gives a range of plausible values: $[0.224, 0.286]$ (`confint(mod_1)`).
- The **p-value** is the probability of an estimate at least as extreme as ours if there was **no** effect (null hypothesis $H_0 : \beta_1 = 0$).

If $p < 0.05$, we reject the null hypothesis $H_0 : \beta_1 = 0$, so the effect is **statistically significant**.

Here, $p < 2 \cdot 10^{-16}$, so the effect of flipper length on bill length is statistically significant.

(Significant does not mean large or important — always look at the size of the estimate as well.)

2.2 Simple linear regression — how much is explained?

The **coefficient of determination** R^2 is the proportion of the variability of the response explained by the model:

$$R^2 = 1 - \frac{\text{SSR}}{\sum_{t=1}^{T}(y_t - \bar{y})^2} \in [0, 1].$$

- $R^2 = 0$: the model explains nothing (flat line); $R^2 = 1$: all points lie exactly on the line.
- Here, $R^2 = 0.43$: 43 % of the variability in bill length is explained by flipper length — and 57 % is not.

What could explain the rest? Let's have a closer look at the residuals!

Lecture 2: Basics of linear regression

2.1 Introduction — what is statistical modelling?

2.2 Simple linear regression

2.3 Model checking

2.4 Multiple linear regression

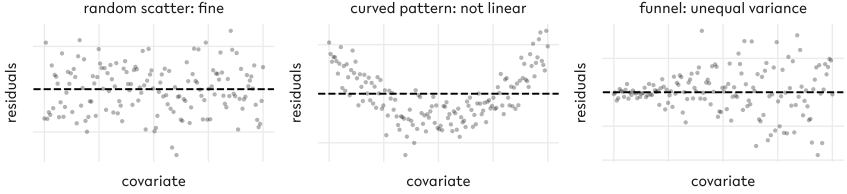
2.5 A non-technical summary

2.3 Model checking — assumptions

Our conclusions are only reliable if the model assumptions are (approximately) met:

- **linearity**: the relationship between x_t and Y_t is linear;
- **independence**: the errors are independent (often violated for time series — Lecture 3!);
- **normality**: the errors are normally distributed;
- **equal variance**: the errors have the same variance for all values of x_t.

We check them using the **residuals** $\hat{\varepsilon}_t = y_t - \hat{y}_t$ (observed minus fitted values):



(simulated data)

2.3 Model checking — in R

Distribution of the residuals — do they follow a normal distribution?

```
res <- residuals(mod_1)
hist(res, breaks = 30, probability = TRUE, col = "grey",
     border = "white", main = "", xlab = "residuals", ylab = "density")
curve(dnorm(x, mean = mean(res), sd = sd(res)), add = TRUE, lwd = 2)
```

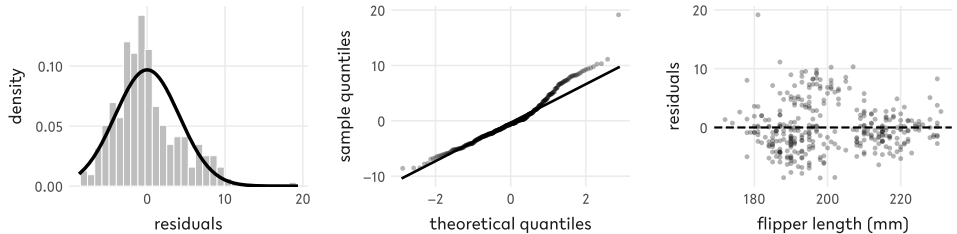
Quantile-quantile (QQ) plot — if the residuals are normal, the points lie on a straight line:

```
qqnorm(res, pch = 20, main = "",
       xlab = "theoretical quantiles", ylab = "sample quantiles")
qqline(res, lwd = 2)
```

Residuals against the covariate — is there a pattern?

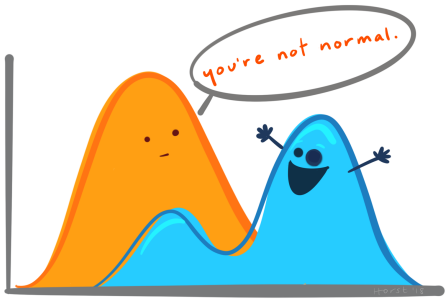
```
x <- penguins$flipper_length_mm[!is.na(penguins$flipper_length_mm)] # without NAs
plot(x = x, y = res, pch = 20, col = adjustcolor("black", alpha.f = 0.3), xlab = "flipper
     length (mm)",
     ylab = "residuals")
abline(h = 0, lty = 2) # horizontal line at 0
```

2.3 Model checking — the penguin model



- The residuals are **right-skewed**, and the QQ plot deviates from the line in the upper tail.
- Doesn't look that good — is there something we haven't taken into account?

2.3 Model checking — what we hope for



Artwork by Allison Horst

If the model assumptions are satisfied, the residuals

- follow a normal distribution;
- show no pattern when plotted against the covariates (or against time);
- have a constant spread.

Residuals that are “not normal” point to something the model has missed.

Lecture 2: Basics of linear regression

2.1 Introduction — what is statistical modelling?

2.2 Simple linear regression

2.3 Model checking

2.4 Multiple linear regression

2.5 A non-technical summary

2.4 Multiple linear regression — the model

Often, **several** covariates affect the response (e.g., flipper length **and** species):

The **multiple linear model** with p covariates is given by

$$\mathbb{E}(Y_t) = \beta_0 + \beta_1 \cdot x_{1,t} + \beta_2 \cdot x_{2,t} + \ldots + \beta_p \cdot x_{p,t},$$

where β_0 is the intercept and $\beta_1, \ldots, \beta_p$ are the regression coefficients. Each β_j is the expected change in Y_t if $x_{j,t}$ increases by 1 **and all other covariates are held fixed**.

Categorical covariates with k categories enter through $k-1$ **indicator variables** (0 or 1); the first category is the **reference**:

species	$\mathbb{1}_{\text{Chinstrap}}$	$\mathbb{1}_{\text{Gentoo}}$
Adélie (reference)	0	0
Chinstrap	1	0
Gentoo	0	1

2.4 Multiple linear regression — estimation in R

$\mathbb{E}(\text{bill length}_t) = \beta_0 + \beta_1 \cdot \text{flipper length}_t + \beta_2 \cdot \mathbb{1}_{\text{species}_t=\text{Chinstrap}} + \beta_3 \cdot \mathbb{1}_{\text{species}_t=\text{Gentoo}}$:

```
mod_2 <- lm(formula = bill_length_mm ~ flipper_length_mm + species, data = penguins)
summary(mod_2)
```

```
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       -2.05859    4.03855  -0.510    0.611
flipper_length_mm  0.21505    0.02123  10.129  < 2e-16 ***
speciesChinstrap   8.78010    0.39912  21.998  < 2e-16 ***
speciesGentoo      2.85689    0.65861   4.338  1.9e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.596 on 338 degrees of freedom
  (2 observations deleted due to missingness)
Multiple R-squared:  0.7758,^^IAdjusted R-squared:  0.7739
```

R creates the indicator variables automatically for categorical variables (factors).

2.4 Multiple linear regression — interpretation

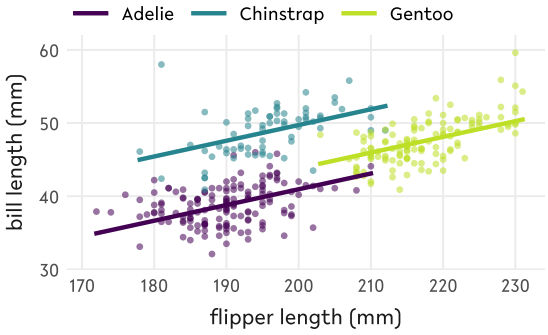
$\mathbb{E}(\text{bill length}_i) = -2.059 + 0.215 \cdot \text{flipper length}_i + 8.780 \cdot \mathbb{1}_{\text{species}_i=\text{Chinstrap}} + 2.857 \cdot \mathbb{1}_{\text{species}_i=\text{Gentoo}}$:

- If the flipper length increases by 1 mm, the expected bill length increases by 0.215 mm (for each species).
- For the same flipper length, Chinstrap penguins have on average 8.780 mm longer bills than Adélie penguins, Gentoo penguins 2.857 mm.
- A Chinstrap penguin with a flipper length of 200 mm:

$$-2.059 + 0.215 \cdot 200 + 8.780 \cdot 1 + 2.857 \cdot 0 = 49.7 \text{ mm}.$$

- All p-values for flipper length and species are below 0.05, so the effects are statistically significant.
- R^2 increased from 0.43 to 0.78.

2.4 Multiple linear regression — plotting the fitted model

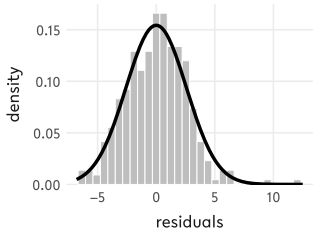


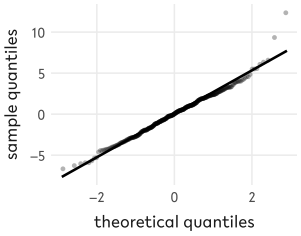
(Colours: `viridis(3, end = 0.9)`, as in Lecture 1.)

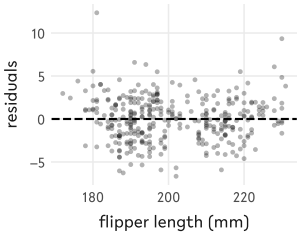
- One regression line per species: the **same slope**, but **different intercepts** (parallel lines).
- The slope is smaller than before (0.215 vs. 0.255): part of the effect of flipper length was in fact a difference **between species**.
- Leaving out an important covariate can distort the estimated effects of the others.

2.4 Multiple linear regression — model checking

The same residual plots as before, now for `mod_2`:







Roughly normal residuals, no obvious pattern — looks much better!

2.4 Multiple linear regression — model selection

Which model is better, `mod_1` or `mod_2`? Model selection using the **Akaike information criterion**:

$$\text{AIC} = \underbrace{-2 \cdot \text{log-likelihood}}_{\text{model fit}} + \underbrace{2 \cdot \text{number of parameters}}_{\text{model complexity}}.$$

- The AIC is a trade-off between **fit** and **complexity**: additional covariates only pay off if they improve the fit sufficiently.
- Select the model with the **lowest** AIC.

```
AIC(mod_1) # 1944.0 (flipper length)
AIC(mod_2) # 1629.1 (flipper length and species)
```

The AIC prefers `mod_2`, the model that includes species.

(R^2 always increases with more covariates — so don't use it to compare models.)

2.4 Multiple linear regression — two continuous covariates

Back to the bike sharing data: $\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \beta_1 \cdot \text{temperature}_t + \beta_2 \cdot \text{wind speed}_t$.

```
bikes <- read.csv("bike_sharing.csv")
mod_3 <- lm(formula = count ~ min_temperature + windspeed, data = bikes)
summary(mod_3)
```

```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     24919.86    3592.54   6.937 4.72e-12 ***
min_temperature  1298.32      47.35  27.420  < 2e-16 ***
windspeed       -2118.64     323.63  -6.546 6.70e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 39390 on 3690 degrees of freedom
  (233 observations deleted due to missingness)
Multiple R-squared:  0.2728,^^IAdjusted R-squared:  0.2724
```

(Temperature in °F, wind speed in mph — converting them is part of the practical.)

2.4 Multiple linear regression — two continuous covariates

$\mathbb{E}(\text{number of bikes}_t) = 24{,}919.9 + 1{,}298.3 \cdot \text{temperature}_t - 2{,}118.6 \cdot \text{wind speed}_t$:

- If the temperature increases by 1 °F (and the wind speed stays the same), the expected number of bikes increases by 1,298.3.
- If the wind speed increases by 1 mph (and the temperature stays the same), the expected number of bikes decreases by 2,118.6.
- At 50 °F and a wind speed of 5 mph:

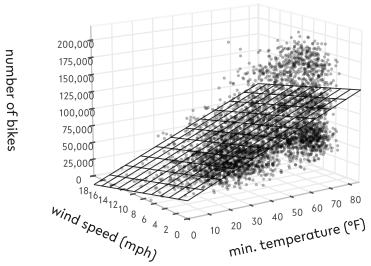
$$24{,}919.9 + 1{,}298.3 \cdot 50 - 2{,}118.6 \cdot 5 = 79{,}243 \text{ bikes}.$$

- The intercept is the expected number of bikes at 0 °F without wind — outside the range of the data.
- Both p-values are below 0.05, so the effects of temperature and wind speed are statistically significant.

2.4 Multiple linear regression — regression surfaces

With two continuous covariates, the regression line becomes a **regression surface**:

```
library(simply3d)
d <- subset(bikes, !is.na(count) & !is.na(windspeed)) # complete days only
p <- simply_scatter(x = d$min_temperature, y = d$windspeed, z = d$count, size = 2,
                    color = "black", xlab = "temperature", ylab = "wind speed",
                    zlab = "number of bikes")
surface(p, expr = predict(mod_3, data.frame(min_temperature = x, windspeed = y)),
        x = d$min_temperature, y = d$windspeed, showscale = FALSE)
```

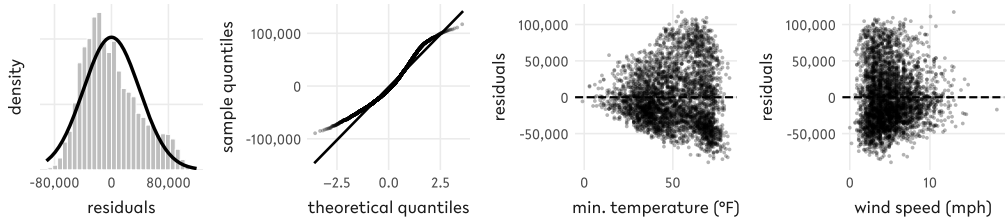


In R, you can rotate and zoom in — try it out!

The surface rises with temperature and falls with wind speed.

2.4 Multiple linear regression — model checking

Residual checks for `mod_3` (same code as before; one plot per covariate):



- The residuals are **right-skewed**: some days have far more bikes than expected.
- Their spread is larger on warm days than on cold days (unequal variance).
- $R^2 = 0.27$: much is unexplained — above all the strong increase over the years. We haven't looked at **time** yet (trend and seasonality in Lecture 3)!

2.4 Multiple linear regression — interaction effects

Does temperature have the same effect on weekdays and weekends? An **interaction effect** allows the effect of one covariate to depend on another:

$$\mathbb{E}(\text{number of bikes}_t) = \beta_0 + \underbrace{\beta_1 \cdot \text{temperature}_t + \beta_2 \cdot \text{weekend}_t}_{\text{main effects}} + \underbrace{\beta_3 \cdot \text{temperature}_t \cdot \text{weekend}_t}_{\text{interaction effect}},$$

where weekend_t is 1 on Saturdays and Sundays and 0 otherwise.

- On weekdays $(\text{weekend}_t = 0)$: the effect of temperature is β_1.
- On weekends $(\text{weekend}_t = 1)$: the effect of temperature is $\beta_1 + \beta_3$.

The interaction effect β_3 is the **additional** effect of temperature on weekends.

2.4 Multiple linear regression — interaction effects in R

```
bikes$weekend <- as.numeric(bikes$day_of_week %in% c(5, 6)) # 1 = Saturday or Sunday
bikes_2025 <- subset(bikes, year == 2025) # 2025, as in Lecture 1
mod_4 <- lm(formula = count ~ max_temperature * weekend, data = bikes_2025)
summary(mod_4)
```

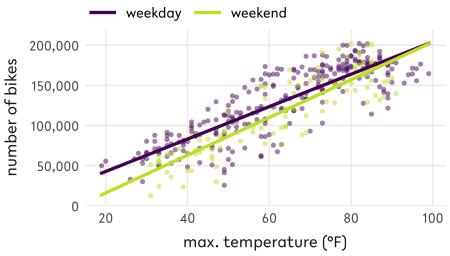
```
Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)               2409.9     5184.6   0.465 0.642350
max_temperature           2017.8       79.4  25.425  < 2e-16 ***
weekend                 -33420.9     9959.3  -3.356 0.000876 ***
max_temperature:weekend    332.5      152.1   2.186 0.029461 *
```

- Weekdays: $+2{,}017.8$ bikes per °F; weekends: $2{,}017.8 + 332.5 = 2{,}350.3$ bikes per °F.
- $p = 0.029 < 0.05$, so the interaction effect is statistically significant: temperature matters **more** on weekends (leisure trips).

(`max_temperature * weekend` is short for `max_temperature + weekend + max_temperature:weekend`.)

2.4 Multiple linear regression — interaction effects in R

```
b <- coef(mod_4)
pal <- viridis(2, end = 0.9) # weekdays, weekends
cols <- adjustcolor(pal[bikes_2025$weekend + 1], alpha.f = 0.5)
plot(x = bikes_2025$max_temperature, y = bikes_2025$count, pch = 20, col = cols,
     xlab = "max. temperature (F)", ylab = "number of bikes")
abline(a = b[1], b = b[2], col = pal[1], lwd = 2) # weekdays
abline(a = b[1] + b[3], b = b[2] + b[4], col = pal[2], lwd = 2) # weekends
legend("topleft", legend = c("weekday", "weekend"), col = pal, lwd = 2)
```



The interaction effect allows for **different slopes** for weekdays and weekends.

Without interaction, the lines would be parallel (as for the penguins).

On cold days, fewer bikes are rented on weekends than on weekdays — on warm days, the difference almost vanishes.

Lecture 2: Basics of linear regression

2.1 Introduction — what is statistical modelling?

2.2 Simple linear regression

2.3 Model checking

2.4 Multiple linear regression

2.5 A non-technical summary

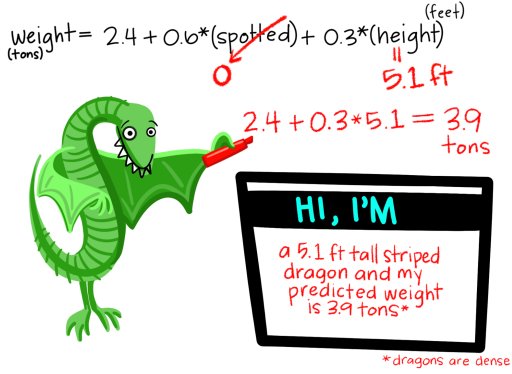
Linear regression — modelling the weight of dragons[2]:



[2] Artwork by Allison Horst, https://allisonhorst.com.

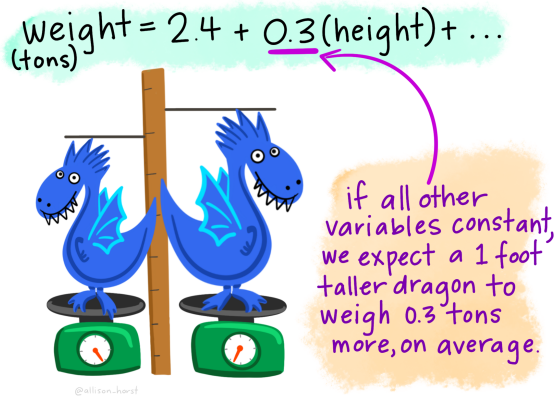
2.5 A non-technical summary

How to **predict** using a fitted multiple linear model?



2.5 A non-technical summary

How to **interpret** the regression coefficient of a continuous covariate?



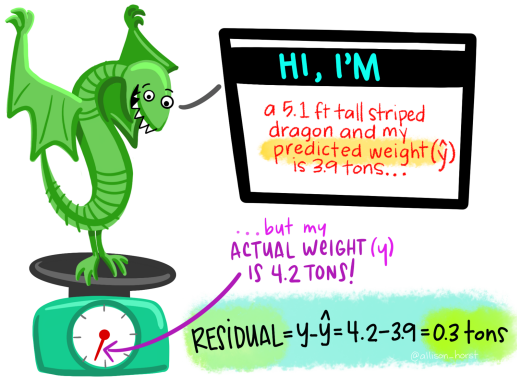
2.5 A non-technical summary

How to **interpret** the regression coefficients of a categorical covariate?



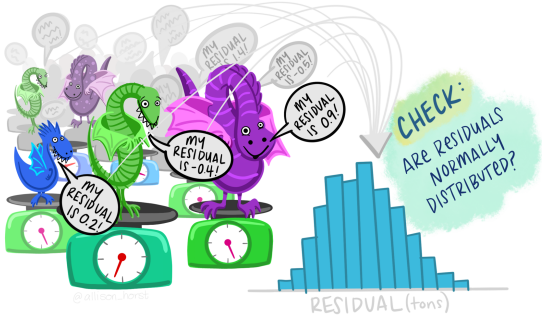
2.5 A non-technical summary

What are **residuals**?



2.5 A non-technical summary

What should the residuals look like?



Lecture 2 — summary

step	question	in R
fit	which line describes the data best?	`mod <- lm(y ~ x1 + x2, data = data)`
interpret	how does Y_t change with x_t?	`summary(mod)`
test	is the effect statistically significant?	`summary(mod)`, `confint(mod)`
check	are the assumptions met?	`residuals(mod)`, `hist()`, `qqnorm()`
select	which model is best?	`AIC(mod_1, mod_2)`
predict	what do we expect for new values?	`predict(mod, newdata = data.frame(...))`

Table: The main steps of a regression analysis.

Response = linear signal + normally distributed noise. Always check the residuals — patterns point to something the model has missed.

Lecture 2 — summary questions

1. What is a statistical model? What are the signal and the noise — and how are they described in a linear model?
2. How are β_0 and β_1 estimated?
3. How do you interpret the estimated slope $\hat{\beta}_1$? What does it mean if the effect is statistically significant — and does a small p-value mean that the effect is large?
4. Which assumptions does the linear model make, and how do you check them using the residuals? Which assumption is most likely violated for time series?
5. How are categorical covariates included in a linear model? Why did the effect of flipper length change when species was added?
6. How do you select between models — and why not using R^2?

Questions? menti.com/al94rhvnrre9.